AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2013

General Mathematics Paper II

Time: 2 hours 20 minutes Marks: 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL Questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER PART a OR PART b OF Q.1.)

Q.1. (Total 5 Marks)

a.

i. Simplify $\frac{x+3}{(x+2)^2} \times \frac{x^2-4}{2x^2+6x}$. (3 Marks)

ii. Find the value of $xy - yz + xz$ when $x = -1$, $y = 1$ and $z = 1$. (2 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.1.)

b. If $x = \sqrt{6} + \sqrt{5}$, then find $x + \frac{1}{x}$ and $x - \frac{1}{x}$. (5 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 5 Marks)

a.

i. Apply remainder theorem to find the remainder if $8x^3 - 2x^2 - 6x + 11$ is divided by $2x - 1$. (3 Marks)

ii. If $x - 3$ is a factor of $x^3 - x^2 + 3x - b$, then find the value of *b*. (2 Marks)

b.

i. Factorize $(x - y)^2 - 12(x - y) + 36$. (3 Marks)

ii. Factorize $ax - bx + ay - by$. (2 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 4 Marks)

a. Find the highest common factor (H.C.F) of $x^4 - y^4$, $x^2 - y^2$ and $(x + y)^2$.

b. Simplify $\frac{x-3}{3x-3} - \frac{x+1}{x-1}$.

Q.4. (Total 5 Marks)

Solve $x - \frac{1}{5} = 3\left(x - \frac{5}{3}\right) - 5x$.

(ATTEMPT EITHER PART a OR PART b OF Q.5.)

Q.5. (Total 5 Marks)

a. Solve $3x^2 = 2(2x-1)$ by using the quadratic formula.

b. The sum of squares of two consecutive odd numbers is 34. Find the smaller number.

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 5 Marks)

a. Do as directed:

i. Write an example of rectangular matrix. (1 Mark)

ii. Write an example of scalar matrix. (1 Mark)

iii. Write a matrix of order 2×3. (1 Mark)

iv. Find the transpose of the matrix $\begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}$. (1 Mark)

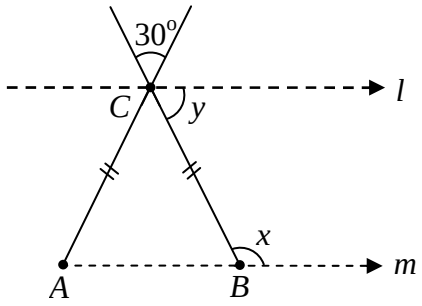
v. Find the additive inverse of the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$. (1 Mark)

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

b. For $\begin{matrix} x+y=5 \\ x-y=1 \end{matrix}$, find the values of x and y by using Cramer's rule. (5 Marks)

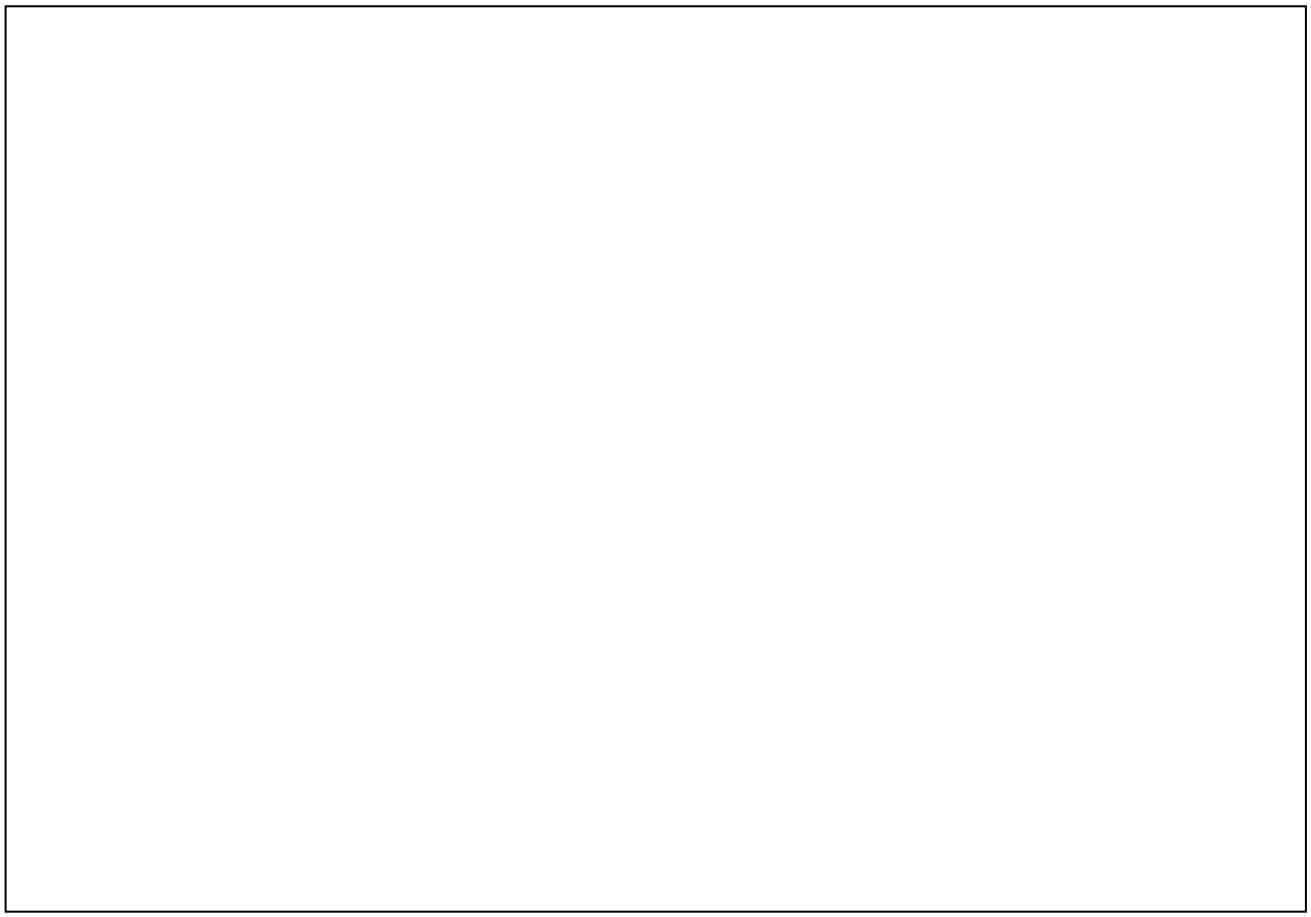
Q.7. (Total 5 Marks)

In the given figure find the values of x and y, where l and m are parallel lines.



Q.8. (Total 4 Marks)

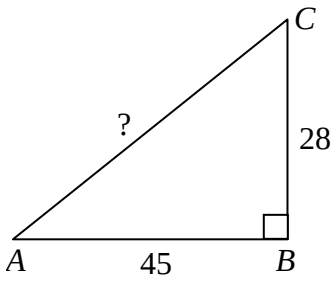
Construct an equilateral triangle *ABC* with each side measuring 3 cm.
Also draw any TWO altitudes of the triangle *ABC*.



Q.9. (Total 4 Marks)

i. Given that the sides of a triangle measure 24, 8 and 25. Is it a right-angled triangle? Justify your answer. (2 Marks)

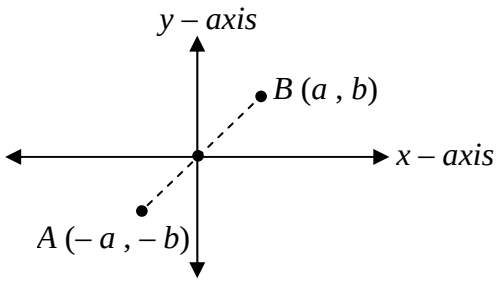
ii. Find the unknown side of the given triangle. (2 Marks)



NOT TO SCALE

Q.10. (Total 3 Marks)

Show that $A\ (-a\ ,\ -b)$ and $B\ (a\ ,\ b)$ are equidistant from origin $O\ (0\ ,\ 0)$.



END OF PAPER

Please use this page for rough work